Research Architect for Coordinated AI Research

Technical white paper with an operational case study from the NWSW project

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Executive summary

Research Architect provides a structured operating framework for research teams working with AI. It converts a reviewed project design into role-specific agent kits, preserves working context across sessions, coordinates evidence and file exchange, and supports verified backup and recovery. The NWSW deployment demonstrates these capabilities in an active research workflow.

Eight people were onboarded through a deployment configured and supported by the framework's creator. Retained project records document hundreds of transfer identities, verified remote deliveries, a live runtime upgrade that preserved 701 exchange files, and an isolated whole-project restore covering more than 123,000 files. The generalized RA 2.1 release subsequently passed 171 offline regression tests. [E01 E02 E03 E04 E08 E13 E22]

Selected deployment and release results

Capability	Documented result
Participant onboarding	8 people onboarded; deployment configured by Brandon Santor.
Exchange records	262 unique transfer IDs in the recorded October 16 inventory.
Live upgrade continuity	701 exchange files totaling 1,655,772,348 bytes preserved and rehashed.
Whole project recovery	123,098 files restored in the original isolated recovery exercise.
Release verification	171 regression tests passed with zero failures, errors or skips.

The case shows how role definition, durable state and verification tools can support continuing human-supervised research. Its practical contribution is a reusable framework whose operational mechanisms were exercised and refined in the field. This paper explains the architecture, describes the NWSW deployment and presents the results that informed RA 2.1.

1 Introduction

AI-assisted research often spans several people, model sessions and collections of files. A useful answer in one chat becomes harder to carry forward when the next session lacks the original scope, evidence, decisions or unresolved work. As a project grows, teams also need a reliable way to identify current versions, distinguish review from delivery, and recover work when a machine or session changes.

Agent literature provides established approaches to combining models, tools and human input. OpenAI's agent-building guide explains foundational components and orchestration choices. AutoGen demonstrates configurable multi-agent conversations and applications. Anthropic's research-system report connects architecture, evaluation and operational engineering in a deployed system. These publications provide useful context for RA's focus on research coordination and continuity. [R1 R2 R3]

Research Architect addresses the organization surrounding the research task. The Architect helps establish questions, methods, assignments, dependencies and review requirements. Its compiler produces individual role kits and coordinated working folders. Operational helpers then preserve state, verify artifacts, support handoffs and maintain recovery records. The chosen AI host supplies the model and available connectors.

RA keeps research responsibility with the people running the project. Its contribution is the operating structure through which their work is assigned, recorded, reviewed and continued. The NWSW case provides concrete evidence of that structure in use, including participant onboarding, actual deliveries, recovery exercises and tested engineering improvements.

2 System architecture

RA organizes research operation around a reviewed design, human-operated roles and persistent project records. Figure 1 shows the logical relationship among these components. Role scopes determine which instructions and project context belong in each kit; configured tools provide the selected evidence and transfer functions.

Project design and role kits

The Architect records the research questions, intended outputs, people, capacity, methods and review assignments. The generator translates that configuration into kits with a named operator, bounded workstream and concrete starting action. The Hub coordinates shared work and releases; Spokes conduct assigned workstreams; the Ingestor organizes incoming material and technical checks.

Persistent operating state

Each session resolves the designated current checkpoint and loads its governing records. New state is published as a successor snapshot, preserving the preceding version. A routed dictionary points the Hub to original evidence and prior decisions. Multipart restoration carries growing context across bounded outputs while preserving source text and repeating governing controls. [E16 E17 E18]

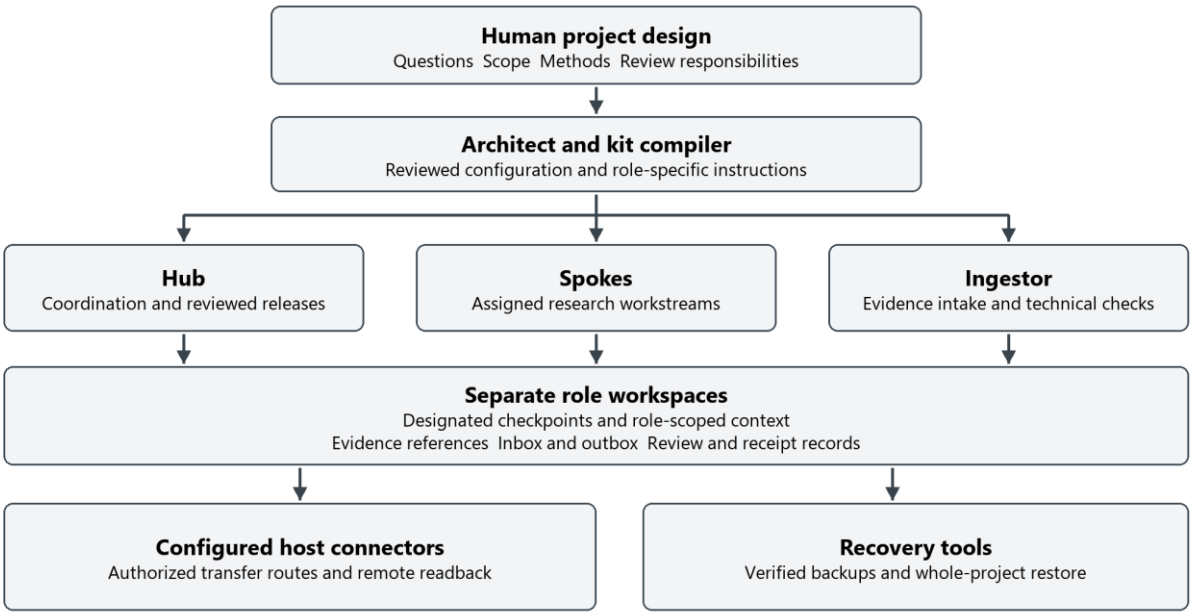


Figure 1 Research Architect components and operating flow

Evidence exchange and recovery

Local helpers prepare and verify artifacts, routes and receipts. Authorized host connectors handle the configured remote transfers during active sessions. Review, delivery and acknowledgment retain separate records. Backup tools capture selected project contents and downloaded Drive artifacts, verify copied bytes, exercise restore and manage explicit rolling retention. [E19 E20]

3 NWSW deployment

The NWSW project supplied an operational environment for RA-generated research kits and a coordinating Hub. I configured the framework and supported onboarding eight people. Participants entered the generated workflow while setup and configuration remained centralized through me. This is the deployment model evaluated in the case. [E01]

The Hub maintained incoming and outgoing exchanges, instructions, source routes, checkpoints and verification records as work continued across sessions. Its operating cycle began with current-state resolution and relevant source lookup. The team then worked on assigned tasks, reviewed the resulting material, prepared authorized handoffs and saved updated state. Versioned instructions and delivery receipts carried changes through the same operating structure.

The review examined selected operational records dated October 8 through October 16, 2026. A read-only resolver selected CP-KW16R and reported a consistent manifest with 904 checked files. Original records were retrieved through the current dictionary and pinned in an evidence manifest. The separate NWSW spatial application was outside this assessment. [E10 E18]

Execution receipts support the transfer, preservation and recovery results below. Creator and recipient statements identify reported onboarding and device outcomes. Native and offline tests support the specified implementation checks. The transfer inventory is fixed at its October 16 observation; individual dispatch receipts provide the evidence for verified deliveries. This keeps the results attached to their actual source and scope. [E01 E02 E13 E25]

4 Operational results

File exchange and versioned instructions

The recorded inventory contained 98 incoming and 164 outgoing tickets with 262 unique transfer IDs. Separate receipts show actual remote distribution: the October 8 repair and notices comprised 15 logical transfers and 30 remote objects; the October 13 distribution comprised 10 transfers and 20 remote files. Both events retained verification of the delivered objects. [E02 E03 E04]

A versioned method instruction reached seven participant routes on October 9. Fourteen retained readbacks cover seven messages and seven addressed tickets. The source recheck verified byte lengths, hashes, names and destination parents. Each instruction body was 4,054 bytes. The instruction register preserves version and supersession history, and a subsequent submission reported retaining and applying the instruction. [E05 E11 E27]

Continuity during a live upgrade

An authorized October 13 Hub upgrade preserved and rehashed 701 exchange files totaling 1,655,772,348 bytes. The preceding exchange store and pointer remained unchanged. Normal resume assembled 183,001 characters, and the accompanying activation validation recorded seven full-context parts. The post-activation exercise passed. [E22 E28]

The verification combined a complete file rehash with inventory checks. Nineteen pages had been validated before activation; one actual CLI page was rerun afterward while the exact inventory snapshot was rechecked. This demonstrated continuity of accumulated operational history through a runtime change. [E22]

Whole project backup and restoration

The October 11 backup captured 744 source files across 48 included Drive folders, producing 748 downloaded or exported artifacts. The completion record also reports 113,813 local files and captured dependencies. Before-and-after enumeration found the captured source set unchanged during the capture window. [E06 E07]

The original isolated restore verified 123,098 files totaling 20,884,075,864 bytes without using live sources. It materialized four dependency links as ordinary directories, verified 38 historical references and completed restored Doctor and context checks. Retained resume and full-context packets support the restoration result. [E08 E29 E30]

Recovery records and their scope

Record	Verified inventory	Use
Original restore exercise	123,098 files; 20,884,075,864 bytes	Historical whole-project recovery result.
Amended current backup	123,090 files; 20,385,912,455 bytes	Current contents following the authorized October 14 backup amendment.
Rolling retention	2 registered snapshots	Explicit maximum of 2 rolling copies.

The current completion records reverified contents after the authorized amendment. The earlier restore remains a historical execution result. The retention registry records two snapshots, and the reusable backup port preserves the existing Hub's current backup controls. [E07 E09 E19 E20]

5 Engineering improvements and release verification

Field operation exposed practical requirements that were carried into the reusable framework: stable current-state selection, publication that behaves correctly on Windows, exact restoration of growing context, reliable retry records and whole-project recovery. The response connected reported behavior to reproducible tests and versioned delivery.

Growing context and Windows setup

The final October 15 context-repair test reconstructed a 466,824-character startup section through five resume parts and eight full-context parts. Its largest part contained 189,188 characters under the unchanged 200,000-character cap. Checks covered exact CRLF and Unicode reconstruction, repeated controls, altered or missing parts and later checkpoint restoration. Four correction files were delivered with remote byte verification. [E12 E21]

A separate Windows path correction passed 16 lifecycle checks, 18 native repair tests and six exchange checks after reproducing the earlier reported path failure. The longest tested path was 245 characters. Four correction files were byte- and destination-verified. On October 16, the participant returned a report of successful activation, a consistent 140-file Doctor check and complete reading of separate 135,189-character context packets. The Hub verified the returned report's bytes and account attribution. [E23 E24 E25]

The report was incorporated into a successor checkpoint that preserved unaffected work, checked 148 routes and recorded zero package resends. This illustrates the support cycle from field report to tested correction, verified distribution and recorded return evidence. [E26]

Reusable framework capabilities

Field experience carried into the reusable package

Capability	Implementation in RA
Current state selection	Designation, identity and hash checks with stale-base rejection.
Reliable publication	Destination-local staging, exclusive publication and readback verification.
Context continuity	Multipart restoration, repeated controls and bounded source lookup.
Exchange operation	Reviewed optional routes, version-specific receipts and guarded lock recovery.
Project recovery	Resumable capture, rolling retention, dependency materialization and standalone restore tools.

RA 2.1.0 distribution revision 2 integrates these capabilities in a separate reusable package. Its provenance record identifies reviewed field inputs and adapted outputs. The evidence-service contract remains 2.0-draft.1, and the deployed Hub retains its own reviewed runtime and history. [E15]

Release verification passed 171 offline regression tests in eleven groups on Windows and Python 3.12.10, with zero failures, errors or skips and no source changes during the run. Coverage includes kit generation, checkpoints, context reconstruction, local exchange, affected evidence services, extracted-package rebuilding and backup recovery. The verified distribution contains 268 files with a complete hash manifest. [E13 E14]

6 Discussion and next phase

The deployment demonstrates a coherent operating workflow across people, sessions and files. Role kits supplied assignments and governing context; checkpoints preserved accumulated work; exchange records made distribution traceable; backup and restore preserved a usable recovery path. These mechanisms were exercised in an active project and then adapted into reusable software.

The engineering results also show the value of retaining operational evidence during change. A live upgrade could be checked against preserved exchange bytes, a context correction could be tested through exact reconstruction, and a returned setup report could be incorporated without resending completed work. This gives subsequent development concrete artifacts and reproducible cases to work from. [E21 E22 E26]

Scope of the demonstrated results

This is a creator-configured deployment case study. The eight-person onboarding count is creator-reported; instruction adoption and remote setup outcomes are recipient-reported. Returned report integrity was verified, while independent device checks and some activation diagnostics remain outside the evidence. Ticket preparation counts and verified deliveries are reported separately. The study does not estimate comparative productivity, cost savings or research-quality gains. These boundaries define the results presented here. [E01 E02 E11 E25 E26]

Next phase of adoption

The next adoption step is for another project lead to configure a new team with the packaged instructions, recording setup time and required support. Deployment work should validate migration and rollback across machines, intended-recipient access and coordinated writer cutover. Longer observation windows can measure transfer completion and support workload. These activities extend the existing participant onboarding and field operation into repeatable deployment by additional configurators.

RA 2.1 remains a development preview. Broader distribution should address hosted identity and authorization where required, supported operating-system behavior and long-term history growth. The current source and evidence records provide a defined basis for that work.

7 Conclusion

Research Architect has supported an eight-person onboarding effort and an active NWSW research workflow with verified exchanges, preserved operating history and whole-project recovery. The deployment produced tested improvements in context restoration, Windows setup, publication and backup tooling, which informed RA 2.1. The resulting framework combines practical field experience with traceable implementation and regression evidence. Its next phase is to make that deployment experience repeatable for additional project leads.

References

Published references

R1 OpenAI. A practical guide to building agents. Technical guide. Accessed October 16 2026.

R2 Qingyun Wu and colleagues. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversations. COLM 2024 research paper. Accessed October 16 2026.

R3 Anthropic. How we built our multi-agent research system. Engineering report June 13 2025. Accessed October 16 2026.

Deployment and implementation records

Record identifiers link the results to the retained Hub evidence and release receipts. Exact source locators, SHA-256 values and detailed verification notes are held in the internal technical record. The case uses the inspection cutoffs stated in Section 3.

E01 Creator onboarding report.

E02 October 16 transfer ticket inventory.

E03 October 8 verified repair dispatch.

E04 October 13 verified repair dispatch.

E05 October 9 method instruction delivery.

E06 October 11 Drive capture stability check.

E07 Current backup completion after authorized purge amendment.

E08 Original October 11 isolated whole project restore.

E09 Current two snapshot retention registry.

E10 Verified checkpoint CP KW16R manifest.

E11 Current instruction register.

E12 October 15 context repair delivery.

E13 RA 2.1 revision 2 regression receipt.

E14 RA 2.1 revision 2 archive verification.

E15 Generalized RA field import provenance.

E16 Current Hub context capacity procedure.

E17 Current Hub checkpoint procedure.

E18 Current Hub operating dictionary.

E19 RA generic backup and retention implementation.

E20 RA generic backup retention implementation.

E21 October 15 final native Windows context repair fixture result.

E22 October 13 live Hub post activation exercise.

E23 October 15 native Windows path repair fixture result.

E24 October 15 Windows v3 correction verified dispatch.

E25 October 16 returned native setup report and Hub verification.

E26 October 16 successor checkpoint intake final receipt.

E27 October 10 instruction retention and application report intake.

E28 October 13 live Hub activation validation and context budget.

E29 Original isolated restore resume packet.

E30 Original isolated restore full context packet.

Prepared with AI-assisted source review and drafting. The author configured the deployment and supplied the onboarding count.